Exploratory Data Analysis (EDA)

Titanic-Style Passenger Sample Dataset

This beginner-level report explores a synthetic Titanic-style passenger dataset. It is created for practice and is not the original historical Titanic passenger dataset.

1. Dataset overview

Measure	Value
Number of rows	500
Number of columns	10
Missing age values	90
Missing embarked values	10
Overall survival rate	49.0%
Median age after cleaning	30.9

2. Data preparation

The age column contained 90 missing values and the embarked column contained 10. Missing ages were filled using the median. Missing embarkation values were filled using the most common category.

3. Visual exploration and observations

Survival count: The bar chart compares the number of passengers in the two survival groups.

Survival by sex: The sample survival rates are 80.8% for female passengers and 31.6% for male passengers.

Survival by class: The sample survival rates are 53.3% in first class, 50.9% in second class and 46.3% in third class.

Age distribution: The histogram shows how ages are spread across the sample.

Fare by class: The boxplot compares fare values across passenger classes and helps show unusual values.

Correlation heatmap: The heatmap summarises relationships between numerical columns. Correlation does not prove cause and effect.

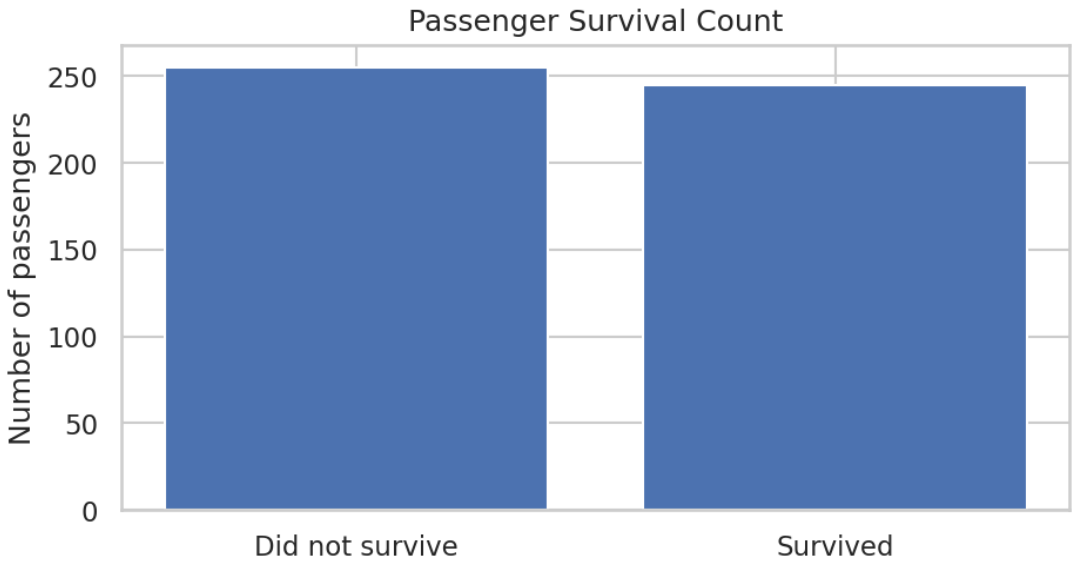


Figure 1. Survival count

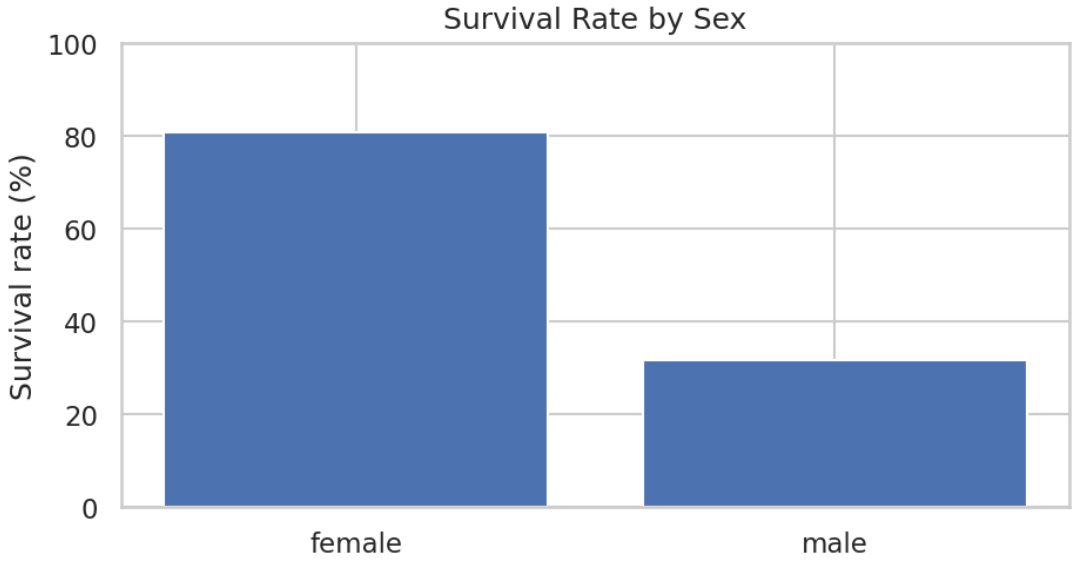


Figure 2. Survival rate by sex



Figure 3. Survival rate by class

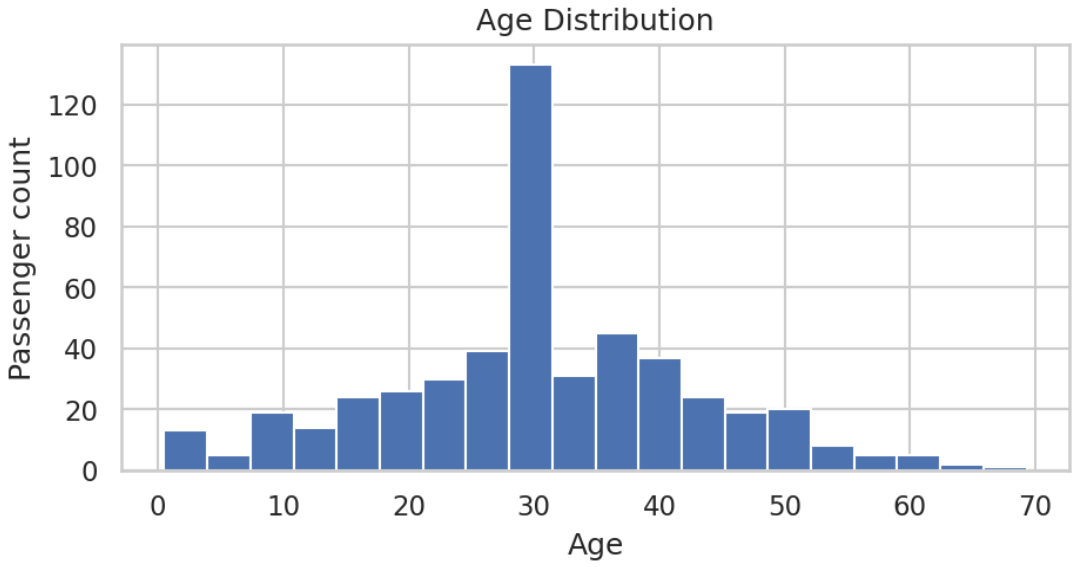


Figure 4. Age distribution

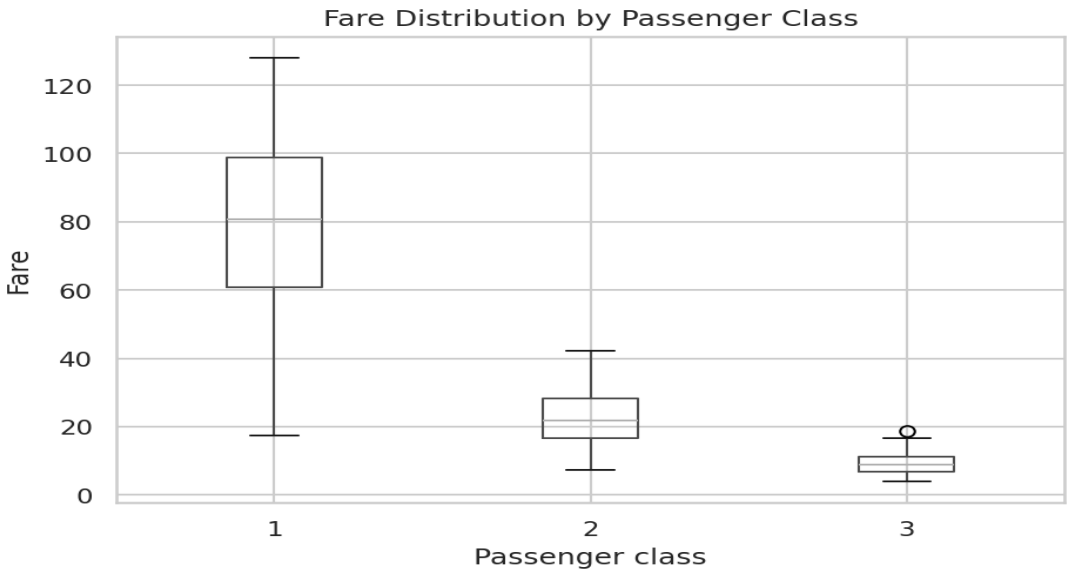


Figure 5. Fare by class

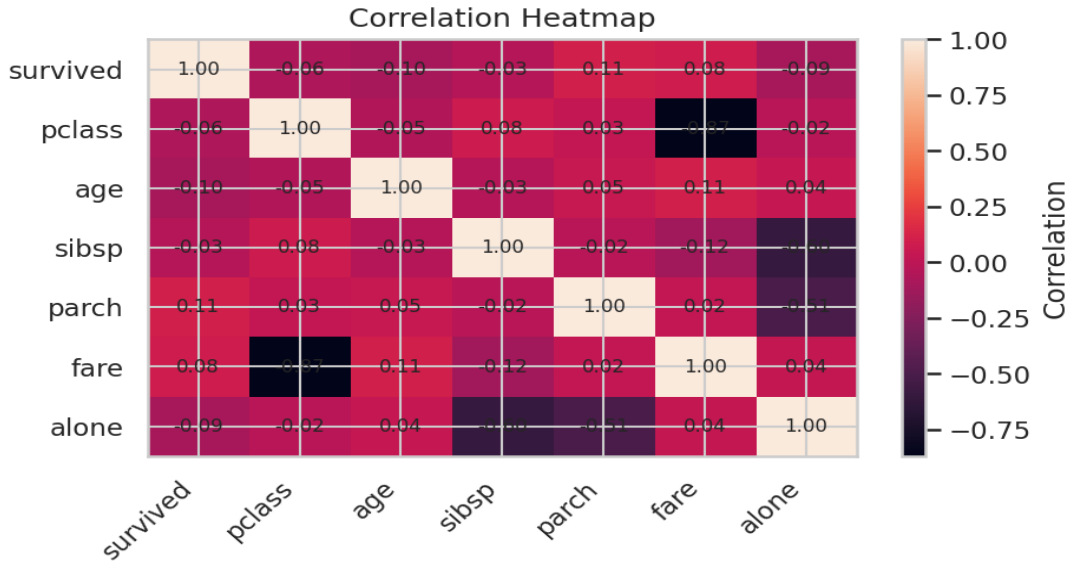


Figure 6. Correlation heatmap

4. Summary of findings

The analysis demonstrates the use of descriptive statistics, missing-value handling and visualisation. In this generated sample, survival rates vary by sex and passenger class. Fare values also vary across classes. These findings apply only to this synthetic teaching dataset.

Tools used: Python, Pandas, Matplotlib and Seaborn.